stabilizer of x in G. Let U be an irreducible representation of G_x. Then we define a representation $V_{(O,U)}$ of $G \ltimes A$ as follows.

As a representation of G, we set

$$V_{(O,x,U)} = \mathrm{Ind}_{G_x}^G U = \{f : G \to U | f(hg) = hf(g), h \in G_x\}.$$

Next, we introduce an additional action of A on this space by $(af)(g) = x(g(a))f(g)$. Then it is easy to check that these two actions combine into an action of $G \ltimes A$. Also, it is clear that this representation does not really depend on the choice of x, in the following sense. Let $x, y \in O$ and $g \in G$ be such that $gx = y$, and let $g(U)$ be the representation of G_y obtained from the representation U of G_x by the action of g. Then $V_{(O,x,U)}$ is (naturally) isomorphic to $V_{(O,y,g(U))}$. Thus we will denote $V_{(O,x,U)}$ by $V_{(O,U)}$ (remembering, however, that x has been fixed).

Theorem 5.27.1. *(i) The representations $V_{(O,U)}$ are irreducible.*

(ii) They are pairwise nonisomorphic.

(iii) They form a complete set of irreducible representations of $G \ltimes A$.

(iv) The character of $V = V_{(O,U)}$ is given by the Frobenius-type formula

$$\chi_V(a, g) = \frac{1}{|G_x|} \sum_{h \in G : hgh^{-1} \in G_x} x(h(a))\chi_U(hgh^{-1}).$$

Proof. (i) Let us decompose $V = V_{(O,U)}$ as an A-module. Then we get

$$V = \bigoplus_{y \in O} V_y,$$

where $V_y = \{v \in V_{(O,U)} | av = y(a)v, a \in A\}$. (Equivalently, $V_y = \{v \in V_{(O,U)} | v(g) = 0$ unless $gy = x\}$.) So if $W \subset V$ is a subrepresentation, then $W = \bigoplus_{y \in O} W_y$, where $W_y \subset V_y$. Now, V_y is a representation of G_y, which goes to U under any isomorphism $G_y \to G_x$ determined by $g \in G$ mapping x to y. Hence, V_y is irreducible over G_y, so $W_y = 0$ or $W_y = V_y$ for each y. Also, if $hy = z$, then $hW_y = W_z$, so either $W_y = 0$ for all y or $W_y = V_y$ for all y, as desired.